

# CodeReview-Parsa-Ian

Ian

## Smells

### **Duplicate Code - randomValue in Item Spawn Methods**

Spawn methods localized to SpawnHelper, this includes both the value assignment and the location assignment of new Items

### **Duplicate Code - collectIfTouched in Item Classes**

Collection handling and methods are localized to the master Item class.

### **Long Class - Separate out paintComponent**

Removed paintComponent class components from GamePanel class and into it's own Game\_Painting class.

### **Static Variables - Item Counts and Item Limits**

Item counts are now in the update file with a simple if statement. Limits and whatnot will be localized to master Item Class

Parsa

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### **Duplicate Code – Score Saving Logic**

Score saving logic was duplicated between update() and saveScoreOnce().

Refactored by removing duplication and centralizing logic in saveScoreOnce().

### **Data Clump – Player Dimensions**

playerWidth and playerHeight were always used together with identical values.

Replaced both with a single constant PLAYER\_SIZE.

### **Long Constructor – GamePanel Setup**

Constructor handled multiple responsibilities (dependencies, assets, input, loop, etc.).

Refactored using helper methods:

initializeDependencies(), loadAssets(), initializePainterAndState(),

`configurePanel(), setupInputHandling(), startGameLoop()`.

### **Long / Complex Method – keyPressed(...)**

Keyboard logic contained multiple responsibilities in one method.

Split into smaller methods:

`handleKeyPressed(), handleMenuKey(), handleControlKey(),`  
`isGameplayInputAllowed()`.

### **Primitive Obsession / Magic Number – Game Loop Delay**

Hardcoded value 16 used in game loop.

Replaced with named constant `GAME_LOOP_DELAY_MS`.

### **Testing**

Verified with:

`mvn clean compile`

`mvn test "-Dtest=GamePanelStateTest,GamePanel_Test"`

All tests passed successfully.